FOOD ORDERS DATA ANALYSIS AND BATCH ELT PIPELINE

Data Analysis and Batch Data Processing Using Python, Pandas and MySQL

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1. INTRODUCTION

Data plays an important role in understanding business activities and making better decisions. Food ordering platforms generate information about orders, customers, payment methods, delivery times, restaurants and financial transactions. Analysing this information helps identify business patterns and understand operational performance.

This project, **Food Orders Data Analysis and Batch ELT Pipeline**, focuses on analysing a food orders dataset and implementing a batch-based ELT pipeline using Python, Pandas and MySQL.

The project is divided into two major parts.

Part 1 – Data Analysis Using Pandas

In the first part, the food orders dataset is explored, cleaned and analysed using Python and Pandas. Missing values, duplicate records, data types and timestamps are examined. Additional columns are created to calculate delivery time and platform net revenue. Business questions are answered using Pandas operations and visualizations.

Part 2 – Batch ELT Pipeline

In the second part, a batch ELT pipeline is developed. The original CSV data is extracted and loaded into a raw MySQL database. The raw data is retrieved in batches, transformed using Pandas, validated and loaded into a separate analytics database. SQL queries are then used to perform business analysis.

The main purpose of this project is to understand both data analysis and data engineering workflows through a practical implementation.

2. PROJECT OBJECTIVES

The main objectives of this project are:

- To understand the structure and contents of a food orders dataset.
- To identify missing values, duplicate records and data quality issues.
- To clean the dataset without making incorrect assumptions.
- To perform data analysis using Pandas.
- To calculate delivery time and platform net revenue.
- To identify useful business insights from the dataset.
- To develop a batch-based ELT pipeline.
- To store original data safely in a raw MySQL database.
- To transform and validate records using Python and Pandas.
- To load valid transformed records into an analytics database.
- To perform business analysis using SQL queries.
- To understand batch processing, database connectivity and pipeline orchestration.

3. DATASET OVERVIEW

The project uses a food orders dataset named `food_orders.csv`.

The dataset contains information related to food orders, including order details, restaurants, payment information, delivery information and financial components.

Dataset Statistics

Description	Value
Original number of records	1,010
Number of columns	14
Records after cleaning	1,000
Exact duplicate rows removed	10
Records with valid Order Value	960

The dataset contains fields such as:

- Order ID
- Customer and restaurant information
- Order and delivery timestamps
- Order Value
- Delivery Fee
- Commission Fee
- Payment Processing Fee
- Refunds/Chargebacks
- Discounts and Offers
- Payment Method
- Order Channel
- Weather information

The dataset is used to understand order patterns, delivery performance and platform revenue.

4. TECHNOLOGIES USED

Technology	Purpose
Python	Main programming language
Pandas	Data cleaning, transformation and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical data visualization
MySQL	Raw and analytics data storage
MySQL Workbench	Database management and SQL execution
MySQL Connector	Python and MySQL connectivity
Jupyter Notebook	Exploratory data analysis
Visual Studio Code	Project development and execution
Git and GitHub	Version control and project storage

5. PART 1 – DATA ANALYSIS USING PANDAS

The first part of the project focuses on understanding and analysing the food orders dataset using Pandas.

The CSV file is loaded into a Pandas DataFrame. Initial inspection is performed to understand the dataset's dimensions, column names, data types and sample records.

The following operations are performed:

1. Importing the required Python libraries.
2. Loading the CSV dataset.
3. Displaying the first and last records.
4. Checking the number of rows and columns.
5. Understanding column data types.
6. Identifying missing values.
7. Checking exact duplicate rows.
8. Checking repeated Order IDs.
9. Validating timestamps.
10. Performing data cleaning and analysis.

The original dataset contains 1,010 records and 14 columns. After removing exact duplicate rows, 1,000 records remain.

6. DATA QUALITY ASSESSMENT

Data quality assessment is performed before cleaning the dataset. It helps identify problems that may affect the accuracy of the analysis.

6.1 Missing Value Assessment

The following missing values were identified in the original dataset.

Column	Missing Records
Discounts and Offers	242
Payment Processing Fee	51
Delivery Fee	50
Payment Method	50
Order Value	41
Commission Fee	40
Refunds/Chargebacks	40

Discounts and Offers has the highest number of missing values.

6.2 Duplicate Record Assessment

Exact duplicate rows were checked using Pandas.

The original dataset contained 10 exact duplicate rows. These records were removed during cleaning.

Repeated Order IDs were also examined to understand possible duplicate order records.

6.3 Data Type Assessment

The data types of the columns were checked to identify whether numerical and date-related fields were stored correctly.

Timestamp columns were converted into datetime values, and financial columns were converted into numeric values for calculations.

6.4 Timestamp Validation

Order and delivery timestamps were checked to identify invalid values and delivery records where the delivery time occurred before the order time.

No delivery-before-order cases were identified in the original timestamp validation.

7. DATA CLEANING DECISIONS

Data cleaning was performed based on the meaning of each column.

Data Quality Issue	Cleaning Decision	Reason
Exact duplicate rows	Removed 10 records	Prevent duplicate counting
Missing Payment Method	Replaced with Unknown	Preserve records without assuming a payment method
Missing Discounts and Offers	Replaced with Not Specified	Distinguish missing information from an actual discount
Missing financial values	Retained as missing	Avoid creating incorrect financial values
Timestamp columns	Converted to datetime	Support date and time calculations
Financial columns	Converted to numeric	Enable mathematical calculations

Missing financial values were not automatically replaced with zero because a missing value does not always mean that the actual financial amount was zero.

Replacing missing fees or refunds with zero could produce incorrect revenue calculations. Therefore, these values were retained as missing and excluded from calculations that required all financial components.

After cleaning, 1,000 records remained.

8. FEATURE ENGINEERING

Feature engineering involves creating useful additional columns from existing data to support analysis.

The following derived columns were created.

8.1 Delivery_Time_Minutes

This column represents the time taken to deliver an order.

It is calculated using the difference between the delivery timestamp and the order timestamp.

Formula:

Delivery Time = Delivery Timestamp − Order Timestamp

The difference is converted into minutes.

8.2 Order_Date

This column extracts the date from the original order timestamp.

It helps analyse order activity based on dates.

8.3 Order_Hour

This column extracts the hour from the order timestamp.

It helps understand the time of day when orders are placed.

8.4 Platform_Net_Revenue

This column represents the net revenue earned by the platform from the available financial components.

Formula:

Platform Net Revenue = Commission Fee + Delivery Fee − Payment Processing Fee − Refunds/Chargebacks

The revenue is calculated only when all four required financial components are available.

9. BUSINESS ANALYSIS AND FINDINGS

Business analysis is performed to understand customer ordering patterns, delivery performance and platform revenue.

The following findings were obtained from the notebook.

9.1 Order Value Analysis

Metric	Result
Valid Order Value records	960
Total Order Value	1,013,415.00
Mean Order Value	1,055.64
Median Order Value	1,040.00

The mean represents the average order value, while the median represents the middle value of the valid order values.

9.2 Payment Method Analysis

Most-used payment method: Cash on Delivery

Number of orders: 336

This shows that Cash on Delivery was the most frequently used payment method in the dataset.

9.3 Order Channel Analysis

Most-used order channel: Mobile App

Number of orders: 686

This indicates that the Mobile App was the most frequently used ordering channel.

9.4 Weather Analysis

Most frequent weather condition: Sunny

Number of orders: 402

This identifies Sunny as the most frequently recorded weather condition among the orders.

9.5 Delivery Time Analysis

Metric	Result
Average Delivery Time	73.58 minutes
Median Delivery Time	74 minutes
Delivery Records Analysed	1,000

The average and median values provide an understanding of delivery performance.

9.6 Platform Net Revenue Analysis

Metric	Result
Total Platform Net Revenue	80,877.00
Records Included	828
Records Excluded	172

A total of 828 records were included because all four financial components required for the revenue calculation were available.

The remaining 172 records were excluded from the calculation because one or more required financial components were missing.

9.7 Top Five Restaurants by Order Count

Rank	Restaurant ID	Number of Orders
1	R2317	6
2	R2726	5
3	R2523	5
4	R2083	5
5	R2016	5

These results identify the five restaurants with the highest order counts in the analysed dataset.

10. DATA VISUALIZATIONS

Data visualizations help understand patterns and relationships in the dataset more easily.

The visualizations created in the notebook are used to support the analysis of food orders and their business characteristics.

The purpose of visualization in this project is to:

- Understand the distribution of numerical values.
- Identify patterns in order-related data.
- Compare categories such as payment methods and order channels.
- Examine delivery-related information.
- Present analytical findings in an understandable way.

The visualizations are used along with numerical calculations rather than as a replacement for them.

11. DATA VALIDATION

Validation is performed to check whether the cleaned data and calculated values are correct.

The following checks were performed:

- Original and cleaned record counts were compared.
- Duplicate rows were checked after cleaning.
- Timestamp columns were checked for valid conversion.
- Financial columns were checked for numeric conversion.
- Delivery time calculations were verified.
- Platform net revenue calculations were checked.
- Sample delivery duration calculations were compared.
- Sample revenue calculations were verified.

Five sample delivery duration checks and five sample revenue checks matched their expected calculations.

The final cleaned dataset contains 1,000 records.

12. PART 2 – BATCH ELT PIPELINE

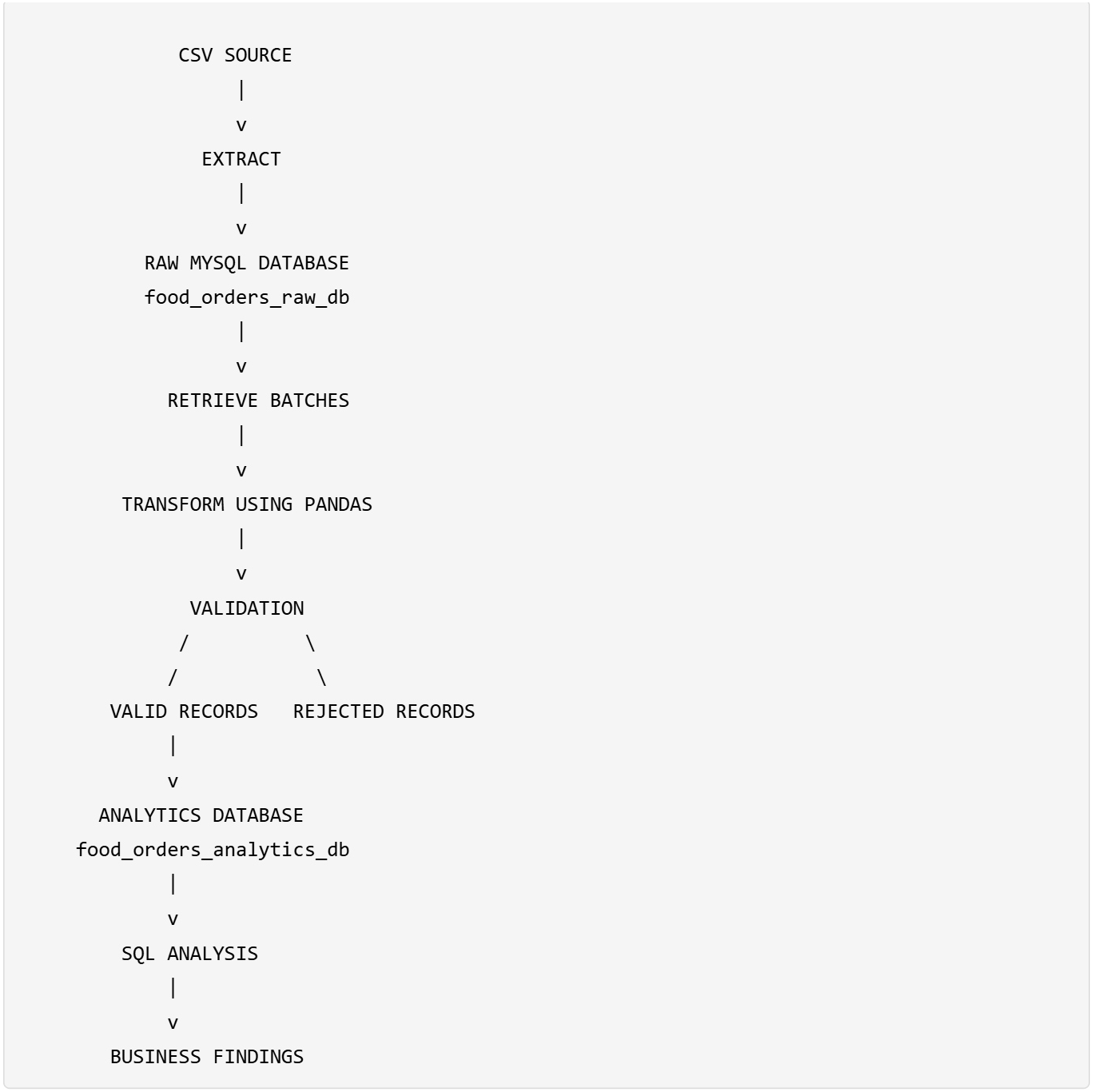
The second part of the project focuses on implementing a batch ELT pipeline using Python, Pandas and MySQL.

ELT stands for:

- **Extract:** Collect data from the source.
- **Load:** Load the original data into the raw database.
- **Transform:** Retrieve the raw data and perform transformations before loading the cleaned data into the analytics database.

Unlike the traditional ETL approach, where transformation happens before loading into the target database, this project first stores the original data in a raw database and then performs the transformation.

12.1 ELT Workflow



This workflow separates original data storage from transformed analytical data.

13. EXTRACTION PROCESS

The extraction process reads the food orders CSV file and prepares the source records for batch processing.

The source dataset contains 1,010 records.

The batch size is configured as 100 records per batch.

Extraction Summary

Description	Result
Source Records	1,010
Batch Size	100
Total Extraction Batches	11

The extraction process divides the dataset into 11 batches. The first 10 batches contain 100 records each, and the final batch contains 10 records.

Batch processing makes it possible to handle the dataset in smaller groups instead of processing all records as a single database operation.

14. RAW DATA LOADING INTO MYSQL

After extraction, the original records are loaded into a MySQL database.

Database Name: `food_orders_raw_db`

Table Name: `raw_food_orders`

The purpose of the raw database is to preserve the source data before transformation.

The raw data is retained with its original values, including missing values and duplicate records. Metadata columns are also maintained for batch loading and tracking.

Raw Loading Summary

Description	Result
Database	food_orders_raw_db
Table	raw_food_orders
Records Loaded	1,010
Loading Batches	11

The raw table provides a stored copy of the source data and allows the transformation process to work from the database rather than directly modifying the original CSV file.

15. RETRIEVAL OF RAW DATA

After loading the source data into MySQL, the raw records are retrieved for transformation.

The retrieval process uses batch-based reading with `fetchmany()` .

Instead of retrieving all records in one operation, records are fetched in groups according to the configured batch size.

Retrieval Summary

Description	Result
Total Records Retrieved	1,010
Batch Size	100
Retrieval Batches	11

All 1,010 records were successfully retrieved across 11 batches.

16. DATA TRANSFORMATION USING PANDAS

The retrieved raw records are transformed using Pandas.

The transformation process applies the required cleaning rules and prepares the records for analytical storage.

The main operations include:

1. Identifying duplicate records across batches.
2. Replacing missing Payment Method values with `Unknown`.
3. Replacing missing Discounts and Offers values with `Not Specified`.
4. Converting timestamp columns into datetime values.
5. Converting financial columns into numeric values.
6. Calculating Delivery_Time_Minutes.
7. Calculating Platform_Net_Revenue when all required financial components are available.
8. Identifying records that fail the transformation rules.

Duplicate detection is performed across batches using record hashes and a set of previously identified hashes.

This helps identify duplicates even when duplicate records are present in different batches.

17. DATA VALIDATION AND REJECTED RECORDS

After transformation, the records are validated before they are loaded into the analytics database.

Validation checks include:

- Duplicate record detection.
- Timestamp validation.
- Numeric data type validation.
- Delivery duration validation.
- Revenue calculation validation.
- Record-count verification.

Records containing duplicate data are rejected during the transformation and validation process.

Validation Summary

Validation Description	Result
Source Records	1,010
Valid Transformed Records	1,000
Rejected Records	10
Rejection Reason	Duplicate Records
Duplicate Check	PASS
Data Type Check	PASS
Delivery Duration Check	PASS
Revenue Calculation Check	PASS
Overall Validation	PASS

All 10 rejected records were identified as duplicate records.

The remaining 1,000 records passed the implemented validation checks.

Rejected records are maintained separately for inspection rather than being included in the analytics table.

18. LOADING TRANSFORMED DATA INTO ANALYTICS DATABASE

After successful transformation and validation, the valid records are loaded into a separate MySQL database.

Database Name: `food_orders_analytics_db`

Table Name: `cleaned_food_orders`

This database stores the cleaned and transformed records that are ready for business analysis.

Analytics Loading Summary

Description	Result
Analytics Database	food_orders_analytics_db
Analytics Table	cleaned_food_orders
Valid Records Loaded	1,000
Loading Batches	10

The transformed records are inserted in batches.

The database record count is verified after loading to confirm that the expected records are available.

19. PIPELINE ORCHESTRATION

Pipeline orchestration is the process of coordinating all the individual steps in the correct order.

The `elt_batch_pipeline.py` module controls the overall execution of the ELT workflow.

It coordinates:

1. Configuration loading.
2. Source data extraction.
3. Raw data loading.
4. Raw data retrieval.
5. Data transformation.
6. Data validation.
7. Loading valid records into the analytics database.
8. Final record-count and execution checks.

Purpose of Python Modules

Module	Purpose
`config.py`	Stores project and database configuration details.
`extract.py`	Reads the source CSV and prepares batches.
`load_raw.py`	Loads original records into the raw MySQL database.
`retrieve_raw.py`	Retrieves raw records from MySQL in batches.
`transform.py`	Performs data cleaning and transformation using Pandas.
`validate.py`	Checks transformed records and separates rejected records.
`load_transformed.py`	Loads valid transformed records into the analytics database.
`elt_batch_pipeline.py`	Coordinates the complete ELT workflow.
`business_analysis.sql`	Contains SQL queries for business analysis.

Batch Insertion

The `executemany()` method is used for inserting multiple records into MySQL in a single batch operation.

This reduces the need to execute a separate insert statement for every record.

Batch Retrieval

The `fetchmany()` method is used to retrieve records in smaller groups from the database.

Transactions and Rollback

Database transactions help maintain consistency during data loading.

If an error occurs during a transaction, rollback can be used to undo the uncommitted changes and prevent incomplete data from being treated as successfully loaded.

Duplicate Loading Prevention

The pipeline checks whether the raw and analytics tables are already populated before performing another load.

This prevents the same records from being inserted again during a normal rerun.

20. SQL BUSINESS ANALYSIS

After loading the transformed data into the analytics database, SQL queries are used to answer important business questions.

The file `business_analysis.sql` contains six business analysis queries.

Query 1 – Total Order Value

This query calculates the sum of valid Order Value records.

Business Purpose: To understand the total order value represented by the available records.

Query 2 – Average Order Value

This query calculates the average Order Value.

Business Purpose: To understand the average value of an order.

Query 3 – Payment Method with the Highest Number of Orders

This query groups records by Payment Method and identifies the method with the highest order count.

Business Purpose: To understand customer payment preferences.

Query 4 – Order Channel with the Highest Number of Orders

This query groups orders by Order Channel and identifies the channel with the highest number of orders. The query excludes NULL channel values.

Business Purpose: To understand which ordering channel is most frequently used.

Query 5 – Average Delivery Time

This query calculates the average delivery time using `Delivery_Time_Minutes`.

Business Purpose: To understand the average time required to deliver orders.

Query 6 – Total Platform Net Revenue

This query calculates the total Platform Net Revenue from the transformed analytics data.

Verified MySQL Workbench Result:

Metric	Result
Total Platform Net Revenue	80,877.00

The MySQL Workbench execution screenshot confirms this result.

The other five queries are included in the SQL file, but their numerical result values are not included in this report because verified output values are not available in the supplied material.

21. FINAL EXECUTION SUMMARY

The complete ELT pipeline was executed and its results were verified.

Initial Successful Loading

Stage	Records	Batches
Source Extraction	1,010	11
Raw MySQL Loading	1,010	11
Raw Data Retrieval	1,010	11
Transformed Valid Records	1,000	—
Rejected Records	10	—
Analytics MySQL Loading	1,000	10

Later Pipeline Rerun

During a later execution, the pipeline identified that the raw and analytics tables already contained the expected records.

Therefore, it skipped loading the records again.

Execution Detail	Result
Extracted Records	1,010
Raw Table Records	1,010
Retrieved Records	1,010
Valid Transformed Records	1,000
Rejected Records	10
Analytics Table Records	1,000
Raw Records Loaded During Rerun	0
Analytics Records Loaded During Rerun	0
Failed Loading Batches	0
Validation Status	PASS
Overall Pipeline Status	PASS

The zero records loaded during the later rerun does not indicate a failure. The records were already present in the respective database tables, so the pipeline skipped duplicate loading.

Final Record Count Comparison

Data Stage	Record Count
Original Source Dataset	1,010
Raw MySQL Table	1,010
Valid Transformed Records	1,000
Rejected Duplicate Records	10
Analytics MySQL Table	1,000

The raw database preserves all original source records, while the analytics database contains the 1,000 valid transformed records.

22. CHALLENGES FACED

During this project, I faced several challenges while working with data analysis and batch processing.

22.1 Handling Missing Values

One challenge was deciding how to handle missing values. I learned that replacing every missing value with zero is not always correct, especially for financial columns.

22.2 Identifying Duplicate Records Across Batches

Duplicate records may appear in different batches. I used record hashing to identify duplicates across the retrieved batches.

22.3 Preserving Original Data

It was important to keep the original data unchanged in the raw database. This helped maintain a separate copy of the source data before transformation.

22.4 Working with Pandas and MySQL

Connecting Python with MySQL and transferring records between them helped me understand how application code interacts with a database.

22.5 Managing Batch Transactions

Handling records in batches required proper insertion, transaction management and record-count verification.

22.6 Validating Calculated Fields

I needed to verify delivery duration and platform net revenue calculations to ensure that the derived values were correct.

22.7 Preventing Duplicate Loading

During pipeline reruns, I had to ensure that already loaded records were not inserted again. This helped me understand the importance of rerun-safe pipeline execution.

23. LEARNING OUTCOMES

By completing this project, I gained practical knowledge in data analysis and data engineering.

The important learning outcomes are:

- Learned how to load and explore CSV datasets using Pandas.
- Understood how to identify missing values and duplicate records.
- Learned to make cleaning decisions based on the meaning of the data.
- Gained experience in converting data types and working with timestamps.
- Learned to create derived columns for business analysis.
- Understood how to calculate order values, delivery time and platform net revenue.
- Gained practical experience in connecting Python with MySQL.
- Understood the difference between raw data storage and analytics data storage.
- Learned how batch extraction, insertion and retrieval work.
- Understood the ELT architecture and its stages.
- Gained experience in data validation and rejected record handling.
- Learned how SQL queries can be used to answer business questions.
- Understood pipeline orchestration and safe rerun behaviour.

This project helped me connect theoretical concepts with a practical data processing workflow.

24. CONCLUSION

The Food Orders Data Analysis and Batch ELT Pipeline project provided practical experience in both data analysis and data engineering.

In Part 1, I analysed the food orders dataset using Python and Pandas. I performed data quality assessment, handled missing values, removed exact duplicate rows, created derived columns and answered business questions. The analysis identified important findings related to order value, payment methods, order channels, weather conditions, delivery time and platform net revenue.

In Part 2, I implemented a batch-based ELT pipeline using Python and MySQL. The original 1,010 records were retained in the raw database. The data was retrieved in batches, transformed using Pandas and validated before loading into the analytics database. A total of 1,000 valid records were loaded, while 10 duplicate records were rejected.

The pipeline execution and validation completed successfully with an overall status of **PASS**.

Through this project, I gained practical knowledge of Pandas, MySQL, SQL queries, batch processing, data transformation, validation and pipeline orchestration. It helped me understand how raw data can be converted into useful and reliable information for business analysis.

END OF PROJECT REPORT